INCH-POUND
MIL-DTL-606G
25 June 2021
SUPERSEDING
MIL-C-606F
18 December 1987

DETAIL SPECIFICATION

CLOTH, BANNER, RAYON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers one (1) type of rayon banner cloth.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online databasehttps://assist.dla.mil/.

AMSC N/A FSC 8305

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

COLOR ASSOCIATION OF THE UNITED STATES

(Copies of this document are available online at https://www.colorassociation.com.)

FEDERAL TRADE COMMISSION

16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Change
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP5 - Evaluation Procedure for Fabric Hand
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM132 - Test Method for Colorfastness to Drycleaning

(Copies of these documents are available on line at https://www.aatcc.org.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Method for Mass per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard sample. Unless otherwise specified, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Yarn. The warp and filling yarn shall be multifilament cuprammonium rayon. The warp yarn shall be 2-ply for both the warp and filling direction.

3.4.2 Weave. The weave shall be 2/2 right twill.

3.5 Color.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

3.5.1 Color. The cloth shall be as specified in the contract (see 6.2) and shall match the standard sample. Shade designations and cable numbers shall be from the Color Association of the United States. Dyeing shall be performed on the yarns prior to weaving using dyes as appropriate to meet the requirements. The finished dyed cloth shall show no more barré than that is shown by the standard sample. The use of dyes and compounds containing sulfur capable of oxidation to sulfuric acid is prohibited. The cloth shall be "Free" of labile sulfur when tested as specified in 4.5.

3.5.2 Visual shade matching. The color and appearance of the dyed rayon cloth shall match the standard sample (see 6.3) when tested as specified in 4.5.

3.6 Physical requirements. The finished cloth shall conform to the colorfastness and physical requirements listed below in Table I, when tested in accordance with the test methods specified in 4.5.

TABLE I. Colorfastness and physical requirements.

Characteristic	Requirement
Weight, oz./yd^2, (min.)	5.0
Yarns per inch (min.)	
Warp	220
Filling	86
Breaking strength, pounds, (min.)	
Warp direction	125
Filling direction	50
Colorfastness: (min.)	
Crocking (wet and dry)	3.0
Light (20 AFU or 85 kJ/(m^{2}nm) @ 420 nm) 1/	3-4
Dry Cleaning (Color Change and Staining) (3 cycles) (min.)	3-4
pH (range)	5.5-8.5

1/ AFU - AATCC Fading Units

3.7 Finish. The finished cloth shall have a hand and drape equal to or better than a standard sample. The use of resin finishes to impart stiffness to the cloth shall be prohibited unless otherwise specifically approved and authorized by the contracting activity.

3.8 Marking. The face side shall be identified by applying a stamping on the face side of the cloth with the word "FACE" at each end of the roll.

3.9 Width. For Government procurements only, the width of the finished cloths shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

3.10 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put-up full width on a roll as specified in the contract or purchase order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

3.11 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. Each roll in the sample shall be examined yard by yard on the face side for defects in accordance with MIL-STD-3064, Type I.

4.4.1 Roll identification and marking examination. During the yard-by-yard examination each roll in the sample shall be examined for roll identification and marking defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation (barré) as specified in MIL-STD-3064.

4.4.3 Length examination (individual and total yardage). During the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 Material testing. The cloth shall be tested for the characteristics listed in Table II. The methods of testing as specified wherever applicable and as listed in Table II shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be three (3) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. Material tests.

Characteristic	Requirement Reference	Test Method
Fiber identification Cuprammonium rayon	See 3.4.1	AATCC TM20 (see 6.4)
Weave	3.4.2	Visual 1/
Labile sulfur	3.5.1	See 4.6.1
Presence of barré	3.5.1	See 4.6.2
Visual shade matching	3.5.2	See 4.6.2
Weight	Table I	ASTM D3776/D3776M (Opt. C)
Yarns per inch	Table I	ASTM D3775
Breaking strength	Table I	ASTM D5034
Colorfastness: Crocking Light Dry cleaning	Table I	 AATCC TM8 2/ AATCC TM16.3, Opt. 3, 3/ AATCC TM132 4/
pH	Table I	AATCC TM81
Finish	3.7	AATCC EP5

1/ One (1) determination shall be made from each sample and results reported as “pass” or “fail”.
2/ Rated using the AATCC EP8.
3/ Rated using the AATCC EP1.
4/ Rated using the AATCC EP1, and AATCC EP2, Evaluation Procedure for Gray Scale for Staining. Only rayon and cotton multifibers shall be evaluated.

4.6 Methods of testing and inspection.

4.6.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as “pass” or “fail.

4.6.2 Visual shade matching. The color and appearance of the material shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 7.6), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200)K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200)K. The presence of barré shall also be evaluated visually against the standard sample.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Department or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD- ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the manufacturing of various types of flags.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. The specific issue of individual documents referenced
c. When first article is required (see 3.1)
d. Width of cloth required (see 3.9)
e. Length required if other than specified (see 3.10)
f. Conformance inspection (see 4.3)
g. Inspection conditions (see 4.3.1)
h. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples address the contracting activity issuing the invitation for bids or request for proposal.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC Evaluation Procedure 9, Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens.

6.6 Subject term (key word) listing.
Barré
Cuprammonium Rayon
Flags

6.7 Changes from previous issue. Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL
Navy - NU
Air Force – 11

Preparing Activity:
DLA-CT

(Project 8305-2021-016)

Review activities:
Army - IH
Navy - MC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:52Z
Check the source to verify that this is the current version before use.